

Bias and variance are key concepts relevant to all of machine learning, not just deep learning or deep reinforcement learning (figure 5.4). Generally, if you reduce bias you will increase variance, and vice versa (figure 5.5). For example, if we ask you to predict the temperature for tomorrow and the next day, you could give us a specific temperature: "The 2-day temperature forecast is 20.1 C and 20.5 C." This is a high-precision prediction—you've given us a temperature prediction to a tenth of a degree! But you don't have a crystal ball, so your prediction is almost surely going to be systematically off, biased toward whatever your prediction procedure involved. Or you could have told us, "The 2-day temperature forecast is 15–25 C and 18–27 C." In this case, your prediction has a lot of spread, or variance, since you're giving fairly wide ranges, but it has low bias, meaning that you have a good chance of the real temperatures falling in your intervals. This spread might be because your prediction algorithm didn't give undue weight to any of the variables used for prediction, so it's not particularly biased in any direction. Indeed, machine learning models are often *regularized* by imposing a penalty on the magnitude of the parameters during training; i.e., parameters that are significantly bigger or smaller than 0 are penalized. Regularization essentially means modifying your machine learning procedure in a way to mitigate overfitting.

Figure 5.4. The bias-variance tradeoff is a fundamental machine learning concept that says any machine learning model will have some degree of systematic deviation from the true data distribution and some degree of variance. You can try to reduce the variance of your model, but it will always come at the cost of increased bias.